D. Y. PATIL COLLEGE OF ENGINEERING AND TECHNOLOGY, KOLHAPUR

(An Autonomous Institute)

Department of Computer Science and Engineering



A MINI-PROJECT REPORT ON

University Management System

Submitted by:

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1. INTRODUCTION

The University Management System is a desktop application built in Python. It manages students, instructors, courses, enrollments, and grades through a Tkinter GUI. The system supports live Matplotlib charts and exports data to Excel and CSV. It demonstrates OOP, file I/O, modular design, and data visualisation.

2. SYSTEM ARCHITECTURE

The project is split into focused modules and a reusable utils package:

File	Responsibility
models.py	Person (ABC), Student, Instructor
courses.py	Department, Course - assignment of instructors and students
enrollment.py	Grade (weighted scoring), Enrollment
university.py	UniversitySystem - CRUD, search, enrollment, GPA calculation
app.py	Tkinter GUI with tabs: Students, Instructors, Courses, Enrollments, Grades, Charts, Export
config.py	Constants: APP_NAME, GPA_SCALE, GRADE_WEIGHTS, LETTER_GRADE_THRESHOLDS
utils/helpers.py	format_currency(), gpa_label(), percentage_to_gpa()

3. KEY FEATURES

- Student, Instructor, Course, Enrollment management with full input validation.
- GPA calculated on a 10-point scale using credit-hour weighted averages.
- Four live Matplotlib charts embedded inside the GUI window.
- Excel export with styled headers and CSV export.
- Modular package structure with shared config and utils.

4. OOP CONCEPTS DEMONSTRATED

Concept	Application in Project
Abstraction	Person is an ABC, set_age() is abstract, forcing subclass-specific age rules.
Inheritance	Student and Instructor inherit from Person, reusing ID counter and validation.
Encapsulation	Private attributes (__name, __gpa, __salary) exposed only via getters/setters.
Polymorphism	display_info() overridden in Student (shows GPA) and Instructor (shows salary).
Composition	UniversitySystem composes Students, Instructors, Courses, and Enrollments.
Packages	utils/ package with __init__.py re-exports helpers used across the project.

5. GUI SCREENSHOTS

The GUI uses a sidebar + tab layout with a white/grey theme and blue accent.

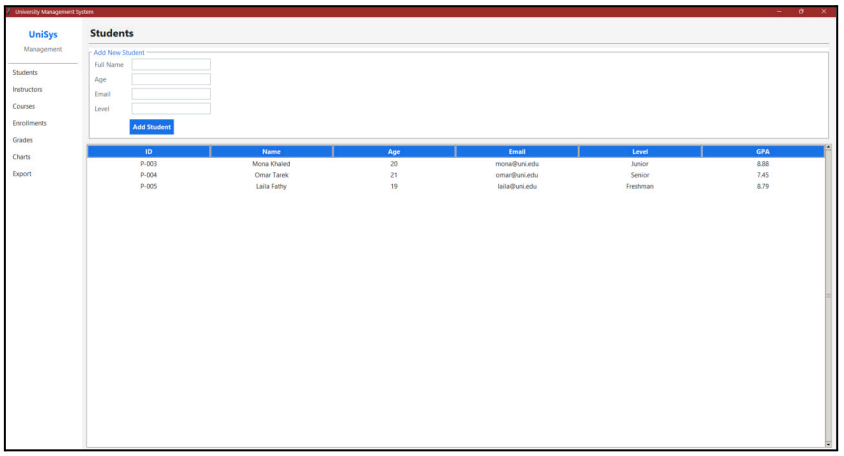


Fig 5.1: Students Tab

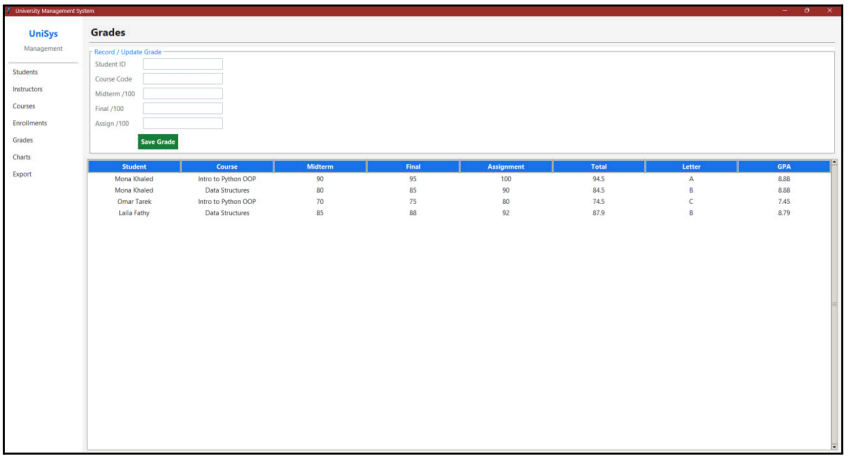


Fig 5.2: Grades Tab

6. MODULES AND PACKAGES

File	Type	Key Exports
app.py	Module	UniversityApp - full Tkinter application
config.py	Module	APP_NAME, GPA_SCALE, GRADE_WEIGHTS
utils/init.py	Package	format_currency, gpa_label, percentage_to_gpa
utils/helpers.py	Module	Utility functions used across modules and GUI
university_system /models.py	Module	Person (ABC), Student, Instructor
university_system /enrollment.py	Module	Grade, Enrollment
university_system /courses.py	Module	Department, Course
university_system /university.py	Module	UniversitySystem

7. CHARTS AND VISUALISATION

The Charts tab embeds Matplotlib figures using FigureCanvasTkAgg. Four chart types are available via buttons:

- GPA per Student - bar chart on 10-point scale.
- Grade Breakdown - grouped bars for Midterm, Final, and Assignment scores.
- Students per Course - pie chart of enrolment share.
- Letter Grade Distribution - bar chart of A/B/C/D/F counts.

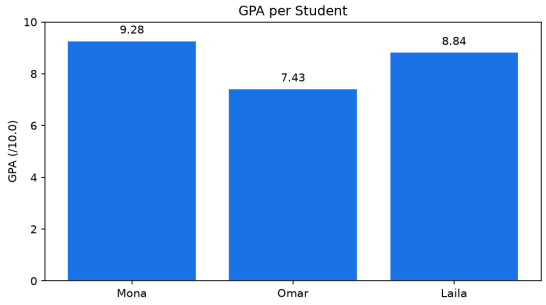


Fig 7.1: GPA per Student

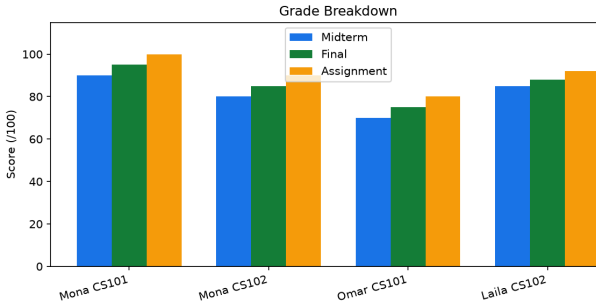


Fig 7.2: Grade Breakdown

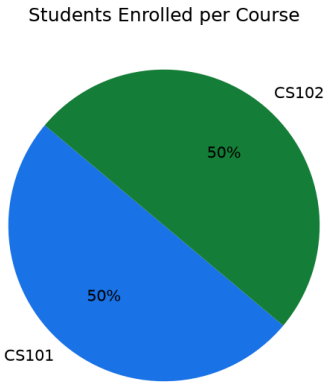


Fig 7.3: Enrolment per Course

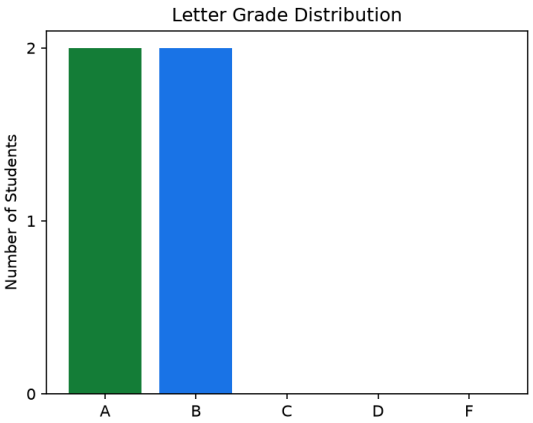


Fig 7.4: Letter Grade Distribution

8. GPA CALCULATION

GPA is computed using a credit-hour weighted average mapped to a 10-point scale:

Total Score = (Midterm x 0.30) + (Final x 0.50) + (Assignments x 0.20)

Weighted Avg = Sum(Score x Credits) / Sum(Credits)

GPA (10.0) = (Weighted Avg / 100) x 10

9. DATA EXPORT

The Export tab provides two format options. Excel uses openpyxl with styled blue headers and auto-fitted columns. CSV uses Python's built-in csv module for lightweight, universal compatibility. Both formats cover Students, Instructors, Courses, and Grades individually. Excel also supports a combined multi-sheet export.

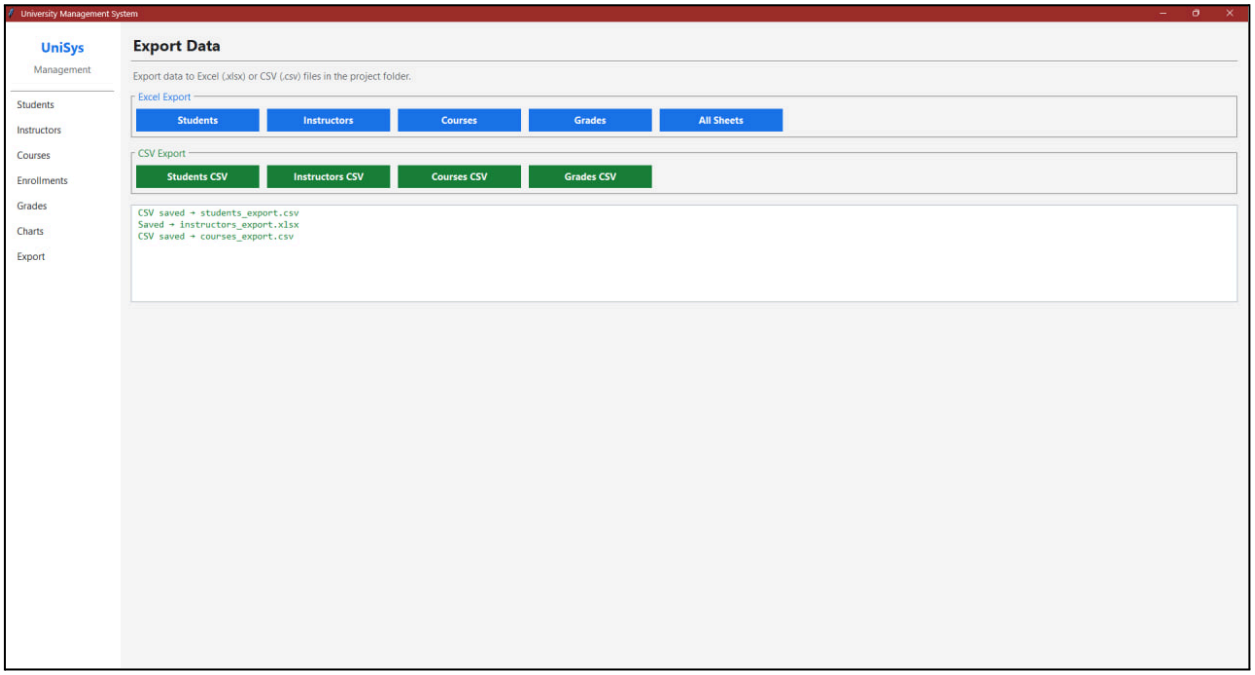


Fig. 9.1: Export Tab - Excel and CSV export buttons with activity log

10. CONCLUSION

The University Management System demonstrates core advanced Python concepts through a practical, fully functional desktop application. OOP principles are applied across a clean modular architecture. The Tkinter GUI provides an intuitive interface for all university operations, Matplotlib delivers embedded live charts, and comprehensive data export in Excel and CSV. Input validation throughout guarantees data integrity at every entry point.